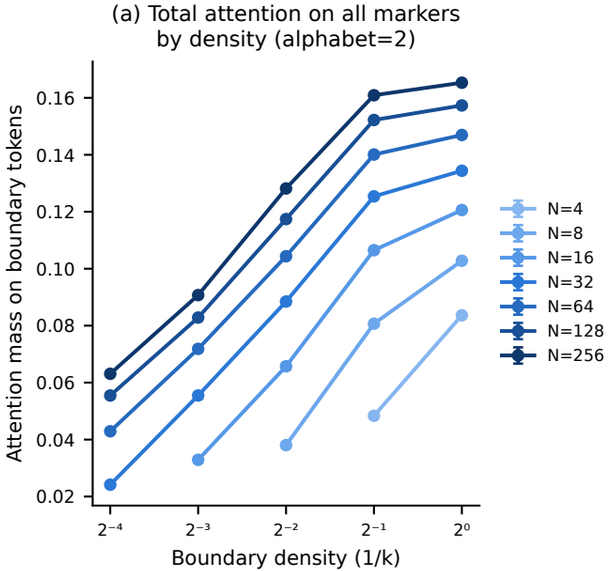
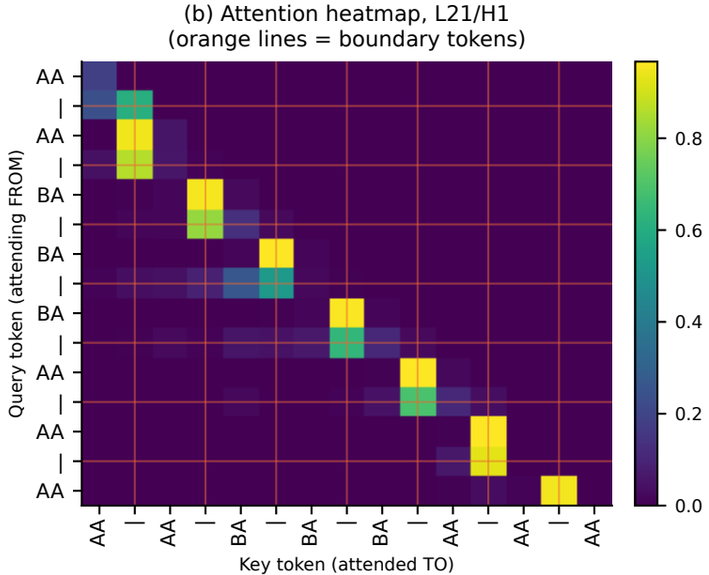
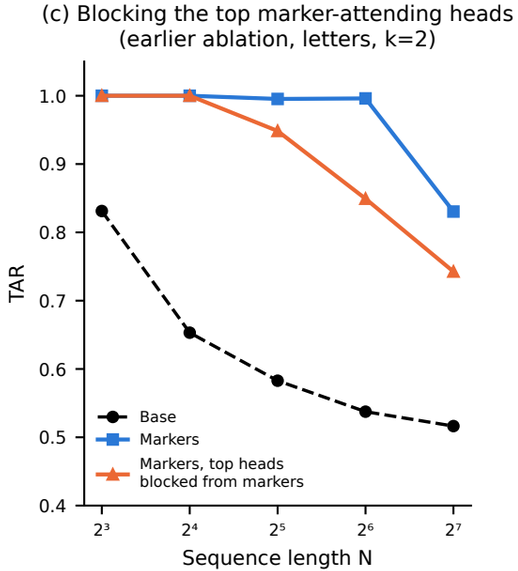
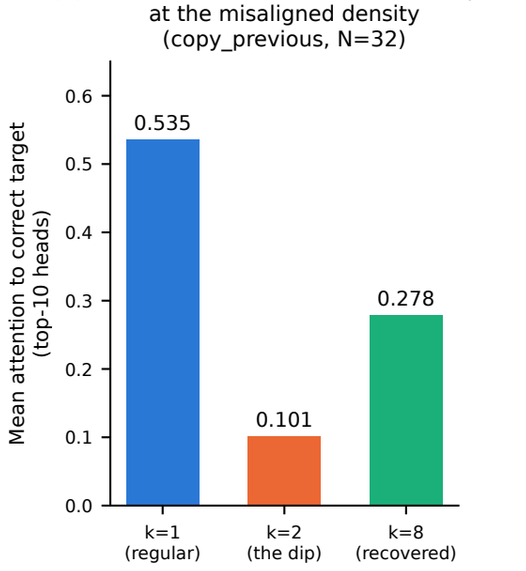

(a) Total attention on all markers
by density (alphabet=2)
Attention mass on boundary tokens
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
2^{-4}
2^{-3}
2^{-2}
2^{-1}
2^{0}
Boundary density (1/k)
N=4
N=8
N=16
N=32
N=64
N=128
N=256


(b) Attention heatmap, L21/H1
(orange lines = boundary tokens)
Query token (attending FROM)
AA
|
AA
|
BA
|
BA
|
BA
|
AA
|
AA
|
AA
Key token (attended TO)
AA
|
AA
|
BA
|
BA
|
BA
|
AA
|
AA
|
AA
0.8
0.6
0.4
0.2
0.0


(c) Blocking the top marker-attending heads
(earlier ablation, letters, k=2)
TAR
1.0
0.9
0.8
0.7
0.6
0.5
0.4
2^{3}
2^{4}
2^{5}
2^{6}
2^{7}
Sequence length N
Base
Markers
Markers, top heads
blocked from markers


(d) Attention to the needed token collapses
at the misaligned density
(copy_previous, N=32)
Mean attention to correct target
(top-10 heads)
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.535
0.101
0.278
k=1
(regular)
k=2
(the dip)
k=8
(recovered)